

KIROV, Russian Federation

WMO: 271990

Lat: 58.567N

Lon: 49.567E

Elev: 158

StdP: 99.44

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-25.8	-32.2	0.2	-29.0	-28.4	0.3	-25.7	7.3	-5.8	6.5	-6.6	1.5	320	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.3	20.3	27.4	19.2	25.7	18.2	21.1	27.4	20.3	26.0	19.3	24.2	2.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.1	23.6	18.2	13.3	22.9	17.3	12.6	22.2	61.9	27.7	58.8	26.3	55.5	24.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.5	4.9	DB	-31.4	31.6	3.9	2.0	-34.2	33.0	-36.5	34.2	-38.7	35.3	-41.5	36.7	(4)
(5)			WB	-31.5	22.4	3.8	1.1	-34.3	23.2	-36.6	23.8	-38.7	24.4	-41.5	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-12.0	-10.7	-4.2	4.0	11.6	16.1	18.7	16.1	10.3	3.2	-4.3	-9.6		
	DBStd	12.08	7.38	7.15	4.61	5.05	5.36	4.50	3.90	4.37	4.25	4.43	5.97	7.40		
	HDD10.0	3239	681	581	439	189	47	5	1	3	46	214	428	606		
	HDD18.3	5584	939	814	697	428	216	96	44	97	241	468	678	864		
	CDD10.0	816	0	0	0	11	96	188	269	192	57	4	0	0		
	CDD18.3	119	0	0	0	0	7	28	55	28	1	0	0	0		
	CDH23.3	988	0	0	0	2	81	226	463	209	8	0	0	0		
	CDH26.7	241	0	0	0	0	14	47	119	60	1	0	0	0		
	WSAvg	2.3	2.6	2.6	2.6	2.4	2.4	2.0	1.8	1.8	1.9	2.4	2.5	2.6		
(15) Precipitation	PrecAvg	612	37	24	28	34	48	66	88	63	69	63	48	44		
	PrecMax	801	70	56	76	82	158	165	254	128	121	183	114	80		
	PrecMin	426	8	1	6	10	12	17	21	6	11	3	12	16		
	PrecStd	106	16	10	17	19	30	39	54	29	27	32	23	18		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.9	7.7	21.7	28.5	30.5	31.6	31.7	24.2	17.2	8.6	4.2		
		MCWB	0.4	0.7	3.7	12.6	18.0	20.9	20.8	21.0	16.2	11.2	7.4	4.0		
	2%	DB	0.5	0.9	5.0	17.4	25.6	28.0	29.7	28.0	21.2	13.4	5.9	1.7		
		MCWB	-0.1	0.2	1.9	10.5	15.8	19.2	20.4	20.1	15.3	10.0	5.2	1.1		
	5%	DB	-0.4	0.1	3.1	14.5	23.1	26.1	27.7	25.4	18.9	11.1	3.8	0.6		
		MCWB	-0.9	-0.5	0.8	8.5	14.5	18.4	19.6	18.7	14.1	8.7	3.2	0.1		
	10%	DB	-2.1	-1.1	1.8	11.8	20.6	24.0	26.0	23.1	16.9	9.4	2.2	-0.2		
		MCWB	-2.6	-1.9	0.1	7.1	13.0	17.0	18.7	17.6	13.2	7.9	1.5	-0.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.1	4.1	13.1	19.2	22.0	22.1	22.0	17.6	12.5	7.5	4.0		
		MCDB	1.0	1.5	6.8	20.3	26.4	28.5	28.2	28.8	21.9	15.4	8.3	4.0		
	2%	WB	0.1	0.3	2.4	11.0	17.0	20.5	21.2	20.7	16.1	10.3	5.3	1.2		
		MCDB	0.4	0.8	4.3	16.0	23.7	26.5	27.3	27.1	19.9	12.4	5.9	1.5		
	5%	WB	-1.0	-0.5	1.3	9.2	15.5	19.2	20.5	19.4	14.7	9.2	3.2	0.2		
		MCDB	-0.5	0.1	2.5	14.0	21.1	24.6	26.3	24.1	18.2	10.8	3.6	0.6		
	10%	WB	-2.6	-1.9	0.5	7.3	14.0	17.9	19.6	18.2	13.4	7.9	1.6	-0.7		
		MCDB	-2.1	-1.2	1.7	11.4	19.2	22.6	24.7	22.3	16.4	9.3	2.1	-0.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	7.1	8.2	9.9	9.3	9.3	8.4	7.3	4.7	4.0	4.9		
		MCDBR	5.0	4.9	7.6	11.9	13.2	11.6	11.9	11.2	10.5	8.0	4.7	3.4		
	5% WB	MCWBR	4.9	4.7	5.5	6.7	6.1	5.1	4.5	4.6	5.5	5.0	4.5	3.4		
		MCWBR	5.0	4.9	6.1	11.2	11.8	10.7	10.6	10.3	9.4	7.1	4.4	3.6		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.309	0.352	0.387	0.371	0.366	0.394	0.396	0.359	0.336	0.298	0.266		
	taud		2.037	2.054	2.007	2.135	2.378	2.421	2.353	2.327	2.408	2.340	2.169	2.012		
	Ebn at Noon		510	684	762	810	861	870	833	799	775	671	495	390		
	Edh at Noon		54	87	122	128	108	105	111	106	84	67	48	39		
(44) All-Sky Solar Radiation	RadAvg		0.45	1.23	2.57	3.89	5.17	5.50	5.48	4.11	2.42	1.01	0.36	0.25		
	RadStd		0.06	0.26	0.34	0.45	0.48	0.45	0.52	0.43	0.31	0.16	0.06	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (6 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-169	N/A	N/A		

Nomenclature: See separate page